

PartB – Parameterized Constructor Lab Report

1. Aim

To develop a Java program using a parameterized constructor, private data members, setter methods, and getter methods to initialize and access book details.

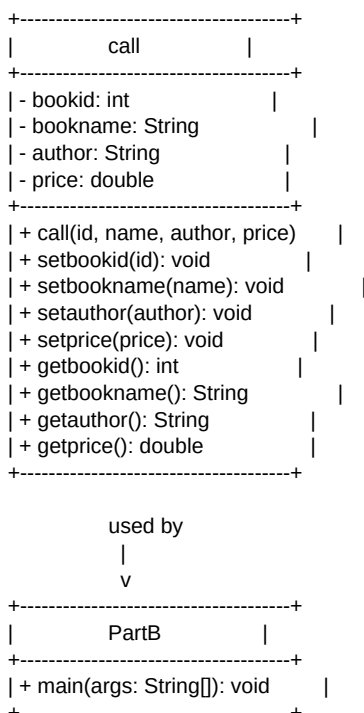
2. Description

The program belongs to the package **tasktwo** and contains a class named **call** and a public class named **PartB**. The call class contains private fields for book ID, book name, author, and price. A parameterized constructor initializes these fields when an object is created. Setter methods modify the values, while getter methods return the values. The main() method creates an object using the constructor and displays the book information.

3. Tools Used

- Programming Language: Java
- Development Environment: Any Java IDE or text editor
- Execution Tool: Java Development Kit (JDK)
- Concepts Used: Parameterized constructor, private data members, encapsulation, setter methods, getter methods, and object creation

4. UML Class Diagram



5. Algorithm

1. Start the program.
2. Define the call class with private fields for book ID, book name, author, and price.
3. Create a parameterized constructor to initialize all four fields.
4. Define setter methods to update the private fields.

5. Define getter methods to retrieve the values of the private fields.
6. Create an object c of the call class using the parameterized constructor.
7. Pass the book ID, book name, author, and price as constructor arguments.
8. Use getter methods to display the book details.
9. Stop the program.

6. Complete Java Source Code

```
package tasktwo;

// Constructor with parameters
class call {
    private int bookid;
    private String bookname;
    private String author;
    private double price;

    public call(int id, String name, String author, double price) {
        this.bookid = id;
        this.bookname = name;
        this.author = author;
        this.price = price;
    }

    void setbookid(int id) {
        bookid = id;
    }

    void setbookname(String name) {
        bookname = name;
    }

    void setauthor(String author) {
        this.author = author;
    }

    void setprice(double price) {
        this.price = price;
    }

    int getbookid() {
        return bookid;
    }

    String getbookname() {
        return bookname;
    }

    String getauthor() {
        return author;
    }

    double getprice() {
        return price;
    }
}

public class PartB {
    public static void main(String[] args) {
        call c = new call(238, "java", "Gosling", 350.50);

        System.out.println("book id is:" + c.getauthor());
        System.out.println("book name is:" + c.getbookname());
        System.out.println("author is: " + c.getauthor());
        System.out.println("price is: " + c.getprice());
    }
}
```

7. Sample Output

```
book id is:Gosling  
book name is:java  
author is: Gosling  
price is: 350.5
```

8. Result

The Java program demonstrates the use of a parameterized constructor and encapsulation. The object is initialized with book details, and getter methods are used to display the stored values. Note: In the supplied code, the first output statement calls `getauthor()` instead of `getbookid()`, so the book ID line displays the author name.